

Brown adipose tissue, whole-body energy expenditure, and thermogenesis in healthy adult men.

Brown adipose tissue (BAT) can be identified by (18)F-fluorodeoxyglucose (FDG)-positron emission tomography (PET) in adult humans. Thirteen healthy male volunteers aged 20-28 years underwent FDG-PET after 2-h cold exposure at 19 °C with light-clothing and intermittently putting their legs on an ice block. When exposed to cold, 6 out of the 13 subjects showed marked FDG uptake into adipose tissue of the supraclavicular and paraspinal regions (BAT-positive group), whereas the remaining seven showed no detectable uptake (BAT-negative group). The BMI and body fat content were similar in the two groups. Under warm conditions at 27 °C, the energy expenditure of the BAT-positive group estimated by indirect calorimetry was $1,446 \pm 97$ kcal/day, being comparable with that of the BAT-negative group ($1,434 \pm 246$ kcal/day). After cold exposure, the energy expenditure increased markedly by 410 ± 293 ($P < 0.05$) and slightly by 42 ± 114 kcal/day ($P = 0.37$) in the BAT-positive and -negative groups, respectively. A positive correlation ($P < 0.05$) was found between the cold-induced rise in energy expenditure and the BAT activity quantified from FDG uptake. After cold exposure, the skin temperature in the supraclavicular region close to BAT deposits dropped by 0.14 °C in the BAT-positive group, whereas it dropped more markedly ($P < 0.01$) by 0.60 °C in the BAT-negative group. The skin temperature drop in other regions apart from BAT deposits was similar in the two groups. These results suggest that BAT is involved in cold-induced increases in whole-body energy expenditure, and, thereby, the control of body temperature and adiposity in adult humans.